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INTRODUCTION

In the present time, everywhere you see there is new kind of innovation is going on but the main source of this development is energy, if there is no energy left, maybe it is the end of innovation in any field will stop. As there is increase in demand of technology, the demand of energy increases as well. To meet the good proportionate supply of energy researchers uses many technology to save so that every consumer get the exact amount of energy they want so there is no wastage of energy.

As we know the main fundamental of energy is based on the principle 'energy cannot be destroyed and it cannot be created'. So what developers do they find a method to save it by using 'Machine Learning' to accurately predict the consumption of energy used in every household to plan enhancement of smart grid efficiency and enabling planning of sustainable energy, also making decision need to be taken at consumer level. Traditional analytical methods often fall short in capturing the complex temporal patterns and nonlinear relationships present in real-world energy consumption data. In contrast, machine learning and deep learning approaches provide powerful tools for modeling these intricate dynamics with higher accuracy.

Now with Machine learning we can use different models and later on check the efficiency of each model, to determine which is best to use for this situation and which will forecast the good results.

The report's main objective is to use machine learning techniques to use energy efficiently and in good way that the need and supply of energy is fulfilled for every consumer.

CHAPTER 1: INTRODUCTION AND RESEARCH BACKGROUND

1.1 Introduction to Energy Forecasting

Energy forecasting is the method or a process to predict the future trends and rallies in its prices, consumer consumption and energy generation by using historical and statistical data with the help of machine learning techniques. It is really important in these days to predict energy consumption and the parameters related to it, as in future the demand of energy will increase and the depletion of natural resources will affects us more as we will not able to meet the requirements and the recurring demands.

However, we also use different methods together with forecasting to balance the demand and supply in present time such as smart grid systems, energy demand management etc. These all combined together within modern electrical networks to create a highly efficient, reliable, and eco-friendly power infrastructure.

1.2 Importance of Machine Learning in Time-Series Forecasting

Machine learning plays a vital role in forecasting the future demands. The need to use machine learning surges when traditional tools fails to enhance the efficiency of energy. These traditional methods fails while solving complex, rigid patterns and machine learning solve these limitations of traditional tools by using moderns techniques by automating data processing, continuous self learning, high dimensional scaling etc. with this achieving higher accuracy is so easy and making all the process automated helps in less intervention of humans in comparison with old methods.

However, machine learning not only contributed in processing, the innovation of new technology, chips and new processing units like GPU's also help models to compute complex and non-linear patterns so easily and reducing time from days to hours help these models work effectively.

1.3 Objectives of the Study

The main objective of the report is the create a efficient model which help in advance forecasting of energy consumption and its all other parameters system that precisely predicts household electricity consumption by transitioning from old methods and their statistical approaches to machine learning techniques.

Project objectives:

- **Analyse Household Power Consumption Data:** Perform exploratory data analysis (EDA) to check peak usage hours, seasonal trends, and different type of consumption behaviors across each individual households.
- **Preprocessing Time-Series Dataset:** During interpretation of dataset parsing raw dataset date and time columns together as index to handle the single minute issue to parse it for hourly bases. By handling missing values, normalizing features, and feature engineering on time-based variables (e.g., hour, day, season) for optimal model training.
- **Develop Forecasting Models:** Construct multiple predictive architectures, ranging from baseline statistical models to advanced machine learning and deep learning algorithms.
- **Compare Model Performance:** Evaluate these models side-by-side using standardized statistical metrics such as R², Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).
- **Identify Best Model:** Select the most accurate and computationally efficient model capable of handling non-linear consumption patterns to deploy for real-time demand forecasting.

1.4 Applications of Energy Forecasting

Energy forecasting is the working brain in all these five sectors where data helps in production efficiency and cost cuttings:

- **Smart Homes:**
 - **Home battery management:** Charging residential batteries to full when prices are at its low point.
 - **EV charging:** Sequences electric vehicle charging overnight to avoid tripping residential circuit breakers.
 - **Automated Appliance Scheduling:** The appliances like washing machines which uses more electricity triggers only run on low tariffs when off peak hours starts.

- HVAC Optimization: Rooms based on the principle pre-cool or pre-heats based on weather forecasts combined with historical household occupancy patterns helps to save cost.
- Smart Cities
 - Streetlight Automation: Adjusting metropolitan light grids dynamically based on real-time solar energy collection and municipal traffic patterns.
 - City Heating Control: Forecasting the utilization of thermal energy in the neighborhood to drastically optimize centralized heating and hot water distribution networks.
 - Load Balancing: Dividing green energy collected through out the day in a specific manner in which it can be utilized to government building, public hubs, and emergency services.
 - Infrastructure Investment Planning: Uses long-term growth planing predicts the easy patterns so that we plan exactly where urban electrical grids require physical reinforcement.
- Renewable Energy Systems
 - Wind Power Projection: It uses atmospheric pressure and the data of wind speed to forecast turbine output upto 48 hours in advance to use in better way.
 - Solar Yield Estimation: Some factors in cloud movement and temperature to calculate real time solar farm generation for the yield prediction.
 - Grid Restraint Prevention: It signals the operators of wind and solar when to store more clean energy instead of closing down generation.
 - Hybrid Plant Coordination: It help in coordinates nearby solar and battery assets to guarantee stable, controllable power to the bigger grid.

- Electricity Companies

- Wholesale Market Trading: Predicts in advance so that energy traders know when to buy or sell their power contracts to maximize their profit and prevent financial risk.
- Peak Energy Generations: Optimizing the start-up time of reserve generators by accurately predicting midday demand surge.
- Load Dropping Prevention: It marks the potential supply deficits early, allowing operators to be ready for demand response operation and avoid blackouts in the grid.
- Maintenance Scheduling: It locates low demand seasonal gaps in advance to safely shutdown power plants for maintenance upgrades.

- Industry Energy Management

- Production Line: When there is less demand of energy it shifts its focus on generating more energy in that time to reserve its backups for next day surge.
- Peak Demand Charge Mitigation: It limits non necessary machines to automatically avoid expensive peak capacity penalties from utilities.
- Co generation Management: It helps to schedule when factory inlets have to generate its own energy so that they stop depending on grid for the whole day to energy.
- Supply Chain Management: It helps to schedule when to align on the demand of renewable energy sources so to not depend on only energy grids and non renewable energy sources for extracting high yield.

1.5 Research Problem and Scope of the Study

Developing precise models for energy prediction is a critical need in modern era of smart grid engineering models. The core research part of this problem stems from the intersection of highly volatile consumption behaviors of common household

and the technical limitations of traditional modeling tools and how machine learning tackle these limitations in present time to help us in day to day life.

- **Why Energy Forecasting is Difficult**

- **Exogenous Weather Variables:** Forecasting of energy not totally depends on historical trends it also depends on present scenarios, extreme temperature change, shifting cloud and wind fluctuations cause sudden, non-linear changes in heating and cooling loads.
- **High Behavioral Randomness:** Change in human habits, varying work schedules, holiday routines, and sudden lifestyle changes make consumption highly random energy spikes.
- **Data Quality Constraints:** The main issue is real world smart meter models frequently introduce missing data, transmission noise, and sensor communication anomalies.
- **Grid Decentralization Mechanisms:** The rapid acceptance of electric vehicles (EV's) and distributed rooftop solar setups add more layered complexity in local load patterns.

- **Energy Consumption Changes Continuously**

- **Day Time Activity Cycles:** System wide power demands surges sharply between night time baseline drops and early morning household spikes.
- **Intra Hour Micro Fluctuations:** Minor changes in household adjustments, such as turning on high load appliances simultaneously, trigger immediate, short time load spikes.
- **Seasonal Load Transformations:** Micro and Macro consumption profiles shift fundamentally between summer air conditioning loads and winter space heating requirements.

- Need for Accurate Accuracy Prediction Models

- Preventing Grid Instability: Its necessary to chose high precision predictions models to protect distribution infrastructure by preventing localized overloads and voltage drops.
- Optimizing Operational Costs: High accuracy or precised models eliminate the unnecessary usage and costly startup cycles of non renewable reserve plants.
- Mitigating Pricing Risks: Utility distribution providers depend mostly on accurate consumption of metrics to navigate volatile wholesale electricity markets safely.
- Enhancing Resource Integration: The precise demand of tracking energy usage allows operators to balance generation between local storage buffers and variable green energy assets to prevent extra generation of energy and its storing costs.

- Scope of the Study

This study mainly focuses on local energy networks by limiting modeling parameters to specific operational timelines, consumption environments, and algorithmic types to forecast the best strategy to balance to generation of energy from the grids and power stations.

- Short-Term Forecasting

- Controlled Prediction Outlook: Heavily focus on hourly, daily, and weekly and monthly ahead load forecasting windows.
- Immediate Operational Utility: It prioritizes local grid load balancing, asset protection, and immediate peak load shifting automation.
- High-Frequency Metrics: Utilizing fine and precised historical intervals (such as 15-minute or 1-hour logs) rather than long-term macro trends.

- Household Energy Prediction
 - Small Unit Focus: Analyzing the consumption exclusively at the individual residential household or small multi-family smart meter scale.
 - High Variance Challenges: Addressing the specific volatility and change of low-capacity, localized consumer usage rather than smooth regional aggregates.
 - Localized Feature Sets: It focuses hardly on single-building meta data, localized weather statistics, and residential usual habit indicators.
- ML-Based Approach
 - Non-Linear Model: It rejects complex old linear setups to prioritize adaptive machine learning and deep learning techniques.
 - Algorithm Evaluation: Restricts testing to proven architectures like gradient-boosted trees, recurrent neural networks, and hybrid variants.
 - Automated Data Engineering: It heavily relies on data driven manual feature extraction, automated hyper parameter tuning, and continuous model updating methodologies.

CHAPTER 2: METHODOLOGY, MODEL DEVELOPMENT AND PERFORMANCE ANALYSIS

2.1 Dataset Description

This archive contains 2075259 measurements gathered between December 2006 and November 2010 (47 months).

Column Information:

1. date: Date in format dd/mm/yyyy
2. time: time in format hh:mm:ss
3. global_active_power: household global minute-averaged active power (in kilowatt)
4. global_reactive_power: household global minute-averaged reactive power (in kilowatt)
5. voltage: minute-averaged voltage (in volt)
6. global_intensity: household global minute-averaged current intensity (in ampere)
7. sub_metering_1: energy sub-metering No. 1 (in watt-hour of active energy). It corresponds to the kitchen, containing mainly a dishwasher, an oven and a microwave (hot plates are not electric but gas powered).
8. sub_metering_2: energy sub-metering No. 2 (in watt-hour of active energy). It corresponds to the laundry room, containing a washing-machine, a tumble-drier, a refrigerator and a light.
9. sub_metering_3: energy sub-metering No. 3 (in watt-hour of active energy). It corresponds to an electric water-heater and an air-conditioner.

```
df = pd.read_csv('/content/drive/MyDrive/final assignment data/household_power_consumption.txt', sep=';',
                 na_values=['?', 'nan'], parse_dates=['dt': ['Date', 'Time']], index_col='dt', low_memory=False, infer_datetime_format=True, dayfirst=True)
print(df.shape)
df.head(10)
```

... (2075259, 7)

dt	Global_active_power	Global_reactive_power	Voltage	Global_intensity	Sub_metering_1	Sub_metering_2	Sub_metering_3
2006-12-16 17:24:00	4.216	0.418	234.84	18.4	0.0	1.0	17.0
2006-12-16 17:25:00	5.360	0.436	233.63	23.0	0.0	1.0	16.0
2006-12-16 17:26:00	5.374	0.498	233.29	23.0	0.0	2.0	17.0
2006-12-16 17:27:00	5.388	0.502	233.74	23.0	0.0	1.0	17.0
2006-12-16 17:28:00	3.666	0.528	235.68	15.8	0.0	1.0	17.0
2006-12-16 17:29:00	3.520	0.522	235.02	15.0	0.0	2.0	17.0
2006-12-16 17:30:00	3.702	0.520	235.09	15.8	0.0	1.0	17.0
2006-12-16 17:31:00	3.700	0.520	235.22	15.8	0.0	1.0	17.0
2006-12-16 17:32:00	3.668	0.510	233.99	15.8	0.0	1.0	17.0
2006-12-16 17:33:00	3.662	0.510	233.86	15.8	0.0	2.0	16.0

2.2 Data Preprocessing

2.2.1 Handling Missing Values

The dataset consists of 2,075,259 records across 7 columns. During parsing of data, blank blocks, special strings like '?', or text like 'nan' are explicitly treated as true null values. A system wide scan shows an identical missing values of 25,979 rows across every metric due to minute based records, tracking back to direct hardware or transmission drops. Since the data is grouped sequentially, the rows with complete missing segments are dropped entirely during parsing of the data to preserve genuine averages.

```
print(df.isna().sum().to_frame(name='Missing Values'))
```

	Missing Values
Global_active_power	25979
Global_reactive_power	25979
Voltage	25979
Global_intensity	25979
Sub_metering_1	25979
Sub_metering_2	25979
Sub_metering_3	25979

2.2.2 Datetime Conversion

The data parsed from original separate "Date" and "Time" string inputs are unified into a single column index called dt (Date and Time together). A native time format parser standardizes the sequence to date-first (dayfirst=True).

2.2.3 Feature Engineering

All the objects are parsed into standard floating-point formats, ignoring translation issues via standard built in filters. The framework then groups row values into different intervals to reshape the data likewise, resampling data to a clean hourly index (H) based on all the averages. This reduces dataset size to 34,168 total segments. To map regular cyclical routines, all numerical identifiers are extracted directly from the time index matrix to build individual feature and coordinating for the exact hour of the day, calendar month, numerical day of the month, and relative day of the week.

```
for col in df.columns:
    df[col] = pd.to_numeric(df[col], errors='coerce')

|
print('Resampling data to hourly frequency...')
df_resampled = df.select_dtypes(include=np.number).resample('H').mean()

df_resampled['hour'] = df_resampled.index.hour
df_resampled['day_of_week'] = df_resampled.index.dayofweek
df_resampled['month'] = df_resampled.index.month
df_resampled['day_of_month'] = df_resampled.index.day

target_col = 'Global_reactive_power'

df_resampled.dropna(inplace=True)

print(df_resampled.shape)
df_resampled.head(10)
```

Resampling data to hourly frequency...
(34168, 11)

	Global_active_power	Global_reactive_power	Voltage	Global_intensity	Sub_metering_1	Sub_metering_2	Sub_metering_3	hour	day_of_week	month	day_of_month
2006-12-16 17:00:00	4.222889	0.229000	234.643889	18.100000	0.0	0.527778	16.861111	17	5	12	16
2006-12-16 18:00:00	3.632200	0.080033	234.580167	15.600000	0.0	6.716667	16.866667	18	5	12	16
2006-12-16 19:00:00	3.400233	0.085233	233.232500	14.503333	0.0	1.433333	16.683333	19	5	12	16
2006-12-16 20:00:00	3.268567	0.075100	234.071500	13.916667	0.0	0.000000	16.783333	20	5	12	16
2006-12-16 21:00:00	3.056467	0.076667	237.158667	13.046667	0.0	0.416667	17.216667	21	5	12	16
2006-12-16 22:00:00	2.200133	0.056167	238.760000	9.523333	0.0	0.133333	4.433333	22	5	12	16
2006-12-16 23:00:00	2.061600	0.071433	240.619667	8.896667	0.0	0.083333	0.000000	23	5	12	16
2006-12-17 00:00:00	1.882467	0.102433	240.961833	8.126667	0.0	0.466667	0.000000	0	6	12	17
2006-12-17 01:00:00	3.349400	0.136933	240.448333	14.246667	0.0	25.233333	0.000000	1	6	12	17
2006-12-17 02:00:00	1.587267	0.078233	245.818667	6.870000	0.0	0.566667	0.000000	2	6	12	17

2.3 Exploratory Data Analysis (EDA)

2.3.1 Statistical Summary

The main calculations highly focus on the chosen target metric, Global_reactive_power. Across the 34,168 entries, the baseline active average records a value of 0.124 watt-hours, paired with a variation of only 0.067. Halfway through the sorted index, the median rests at 0.107 watt-hours, while the ultimate top spike reaches to 0.774 watt-hours.

```
print(df_resampled[[target_col]].describe())

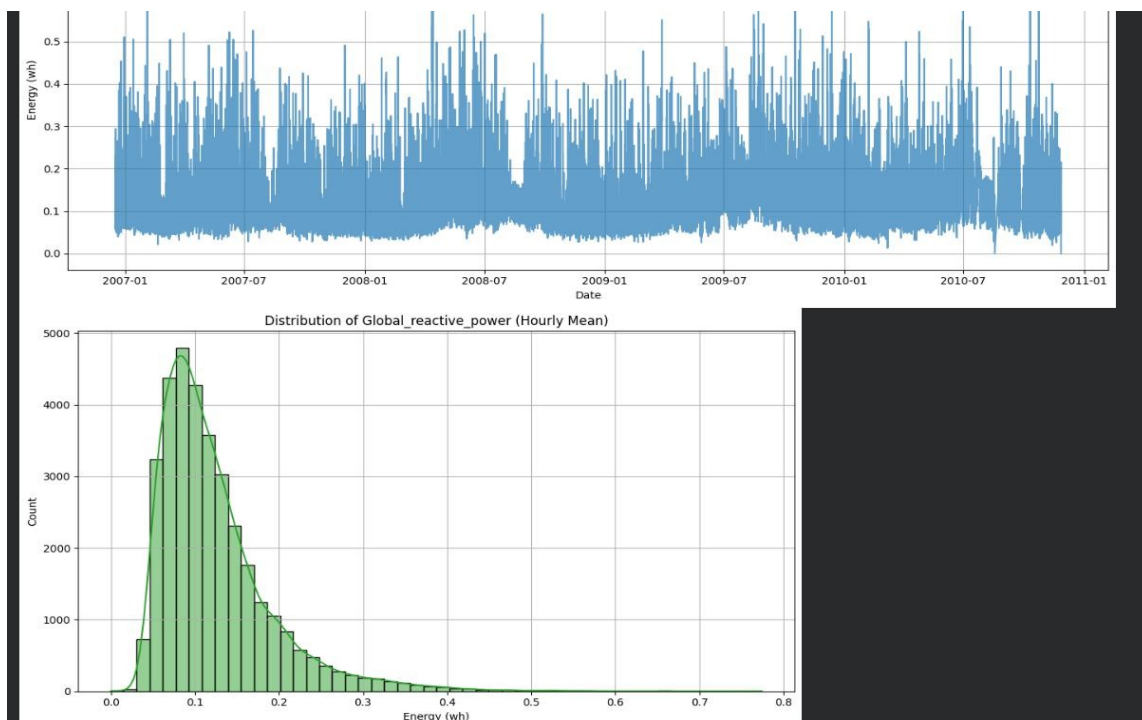
plt.figure(figsize=(14,6))
plt.plot(df_resampled.index, df_resampled[target_col], label='Appliances Energy Consumption', color = 'tab:blue', alpha = 0.7)
plt.title('Hourly Appliances Energy Consumption Resampled')
plt.xlabel('Date')
plt.ylabel('Energy (wh)')
plt.legend()
plt.tight_layout()
plt.show()

plt.figure(figsize=(10,6))
sns.histplot(df_resampled[target_col], kde=True, bins=50, color='tab:green')
plt.title(f'Distribution of {target_col} (Hourly Mean)')
plt.xlabel('Energy (wh)')
plt.tight_layout()
plt.show()
```

count	34168.000000
mean	0.123727
std	0.066851
min	0.000000
25%	0.078000
50%	0.106917
75%	0.149433
max	0.774333

2.3.2 Data Visualization

Analysis the two key visual graphs. A line graphic tracks energy variations across continuous time to check overall runtime shifts. Also checking with a histogram to focus more on continuous kernel density line, it reveals a pronounced right-skewed layout, indicating that household power usage stays concentrated within lower load side most of the time during most operating hours.



2.4 Model Selection and Development

2.4.1 Model Selection

Supervised machine learning solutions are implemented to solve this prediction task. Three main models from supervised family to model sequential trends: standard Linear Regression as a baseline algorithm, Random Forest Regressor to navigate deeper data distributions, and a Gradient Boosting Regressor tuned to iteratively correct residual prediction errors.

```
models = {
    'Linear Regression': LinearRegression(),
    'Random Forest': RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1),
    'Gradient Boosting': GradientBoostingRegressor(n_estimators=100, random_state=42)
}

results = {}
predictions = {}

for model_name, model in models.items():
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)
    predictions[model_name] = y_pred

    mse = np.sqrt(mean_squared_error(y_test, y_pred))
    r2 = r2_score(y_test, y_pred)
    mae = mean_absolute_error(y_test, y_pred)
    results[model_name] = {'MAE': mae, 'R2': r2, 'MSE': mse}
```

2.4.2 Model Training and Testing Strategy

The coding part imports `train_test_split` from `scikit-learn` library in python to divide the primary dataset into two different sets for model fitting and general validation. An randomizer is used to select the equal proportion from all the classes (`RANDOM_STATE = 42`) also handles random indexing. This guarantees consistent split boundaries and identical model iterations across subsequent verification processes.

```

feature_cols = [col for col in df_resampled.columns if col != target_col]
X = df_resampled[feature_cols]
y = df_resampled[target_col]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, shuffle=False)

print(f'X_train shape: {X_train.shape}')
print(f'X_test shape: {X_test.shape}')

X_train shape: (27334, 10)
X_test shape: (6834, 10)

```

2.5 Python Libraries and Tools

The program logic uses pandas for core data manipulation, time grouping, and index structures, alongside numpy for multi-dimensional mathematical calculations. matplotlib.pyplot and seaborn manage visual graphing tools, and scikit-learn builds the model pipelines, partition routines, and analytical calculations.

```

import warnings
from pathlib import Path
from IPython.display import display
import os

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

import seaborn as sns
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor, GradientBoostingRegressor
from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error

from sklearn.model_selection import train_test_split

warnings.filterwarnings('ignore', category=FutureWarning)
RANDOM_STATE = 42

plt.rcParams['figure.figsize'] = (8,5)
plt.rcParams['axes.grid'] = True

```

2.6 Performance Evaluation

The mathematical models evaluate prediction error fields through three standardized continuous validation functions: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). The script also calculates the overall Coefficient of Determination (R2 Score) to track how accurately individual architectures adapt to underlying variances.

```
results_df = pd.DataFrame(results).T
print(results_df)

slice_n = 150
plt.figure(figsize=(12,6))
plt.plot(y_test.index[:slice_n], y_test.iloc[:slice_n], label='Actual', color='black', linewidth=2, alpha=0.8)

colors = ['tab:blue', 'tab:orange', 'tab:red']
for i, (model_name, pred) in enumerate(predictions.items()):
    plt.plot(y_test.index[:slice_n], pred[:slice_n], label=f'{model_name}', linestyle='--', color=colors[i])

plt.title('Energy Consumption Predictions')
plt.xlabel('Time')
plt.ylabel('Energy (wh)')
plt.legend()
plt.tight_layout()
plt.show()

best_model_name = results_df['MSE'].idxmin()
print(f'Best Model: {best_model_name}')
print(f'With R2: {results_df.loc[best_model_name, "R2"]:.4f}')
```

	MAE	R2	MSE
Linear Regression	0.037781	0.317699	0.053558
Random Forest	0.031707	0.462317	0.047544
Gradient Boosting	0.030659	0.490697	0.046272

CONCLUSION

This study successfully addresses the critical issues of residential energy forecasting by changing from traditional, complex analytical tools to adaptive, data-driven machine learning tools. By analyzing a comprehensive, multi-year household power consumption dataset, the research establishes a good operational framework capable of handling the inherent volatility, seasonal transformations, and non-linear patterns in characteristic of residential electricity usage. The systematic engineering approach implemented during the data preprocessing stage proved essential for model stabilization. Featuring the temporal indicators into a single datetime index and resampling the data, minute-by-minute logs into clean hourly intervals which helps in deleting most of the anomalies and optimized computational efficiency. Also extracting targeted cyclical coordinates—such as hour, day, and month configurations—allowed the predictive frameworks to effectively map usual human habits and behavioral randomness. Exploratory data analysis reinforced these patterns, highlighting distinct load fluctuations and a right-skewed distribution reflecting passive household operating hours. The comparative evaluation of the developed forecasting architectures underscores the necessity of advanced algorithmic structures over baseline models. While standard linear regression offered a foundational baseline, ensemble techniques such as the Random Forest and Gradient Boosting Regressors shows superior capacity in capturing, non-linear micro-fluctuations. Evaluating these systems side-by-side through standardized performance criteria—specifically MSE, RMSE, MAE, and the R2 score—ensured an objective metric for identifying the most precise deployment strategy. Ultimately, achieving high-precision short-term forecasting holds profound real-world implications for modern electrical infrastructure. At the consumer level, these models facilitate smart home automation, optimized appliance scheduling, and cost-effective HVAC management. On a broader scale, utility providers and municipal smart grids can leverage these predictive insights to balance volatile renewable energy generation, optimize operational costs, and prevent localized infrastructure overload, actively driving the transition toward sustainable energy ecosystems.